

Practice Problems: Time Series Econometrics

Persistence, Trends, Forecasting, and Dynamic Systems

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How to use this practice set

Structure

The questions become progressively harder:

- Problems 1–3: time-series language, stationarity, and autocorrelation.
- Problems 4–6: AR(1) mechanics, forecasting, and unit roots.
- Problems 7–8: time-series regression, serial correlation, and distributed lags.

Instruction

Attempt each problem before opening the solution. The goal is not only to compute, but to state the maintained assumptions clearly.

Problem 1: Basic time-series language

Let $\{Y_t : t = 1, \dots, T\}$ be a monthly time series for inflation.

- (a) What is the difference between a stochastic process and one observed time-series realization?
- (b) Define the lag operator L and write ΔY_t using L .
- (c) If P_t is a price index, why is $\Delta \log P_t$ often interpreted as an inflation rate?

Solution

Problem 2: Weak stationarity

Consider the process

$$Y_t = \mu + \varepsilon_t + 0.5\varepsilon_{t-1},$$

where $\{\varepsilon_t\}$ is white noise with $\mathbb{E}(\varepsilon_t) = 0$, $\text{Var}(\varepsilon_t) = \sigma^2$, and $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$.

- (a) Compute $\mathbb{E}(Y_t)$.
- (b) Compute $\text{Var}(Y_t)$.
- (c) Compute $\text{Cov}(Y_t, Y_{t-1})$ and $\text{Cov}(Y_t, Y_{t-2})$.
- (d) Is Y_t weakly stationary? Explain.

Solution

Problem 3: Reading autocorrelations

Suppose a researcher computes sample autocorrelations for three variables:

Series	$\hat{\rho}_1$	$\hat{\rho}_2$	$\hat{\rho}_3$
A	0.04	-0.02	0.01
B	0.78	0.60	0.47
C	0.98	0.96	0.94

- (a) Which series looks closest to white noise?
- (b) Which series looks persistent but plausibly stationary?
- (c) Which series raises concern about a unit root or strong trend?
- (d) Why is an ACF pattern only a diagnostic, not a proof?

Solution

Problem 4: AR(1) mean, variance, and ACF

Let

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2),$$

with $|\phi| < 1$.

- (a) Derive the unconditional mean $\mu = \mathbb{E}(Y_t)$.
- (b) Derive the unconditional variance $\text{Var}(Y_t)$.
- (c) Derive $\rho_h = \text{Corr}(Y_t, Y_{t-h})$.
- (d) What happens to persistence as ϕ approaches one from below?

Solution

Problem 5: AR(1) forecasting

Suppose quarterly inflation follows

$$Y_t = 1 + 0.6Y_{t-1} + \varepsilon_t, \quad \mathbb{E}(\varepsilon_{t+h} \mid \mathcal{F}_t) = 0.$$

At date T , the observed inflation rate is $Y_T = 4$.

- (a) Compute the unconditional mean of Y_t .
- (b) Compute the one-step-ahead forecast $\hat{Y}_{T+1|T}$.
- (c) Compute the two-step-ahead forecast $\hat{Y}_{T+2|T}$.
- (d) Explain why the forecast converges to the unconditional mean as the horizon grows.

Solution

Problem 6: Unit root versus trend stationarity

Consider two models:

$$A: Y_t = \alpha + \delta t + u_t, \quad u_t = \rho u_{t-1} + \varepsilon_t, \quad |\rho| < 1,$$

$$B: Y_t = Y_{t-1} + g + \varepsilon_t.$$

- (a) Which model is trend-stationary and which is difference-stationary?
- (b) For each model, what transformation would you use before fitting a stationary ARMA model?
- (c) What is the long-run effect of a one-time shock ε_t in each model?
- (d) Why can confusing A and B lead to misleading empirical conclusions?

Solution

Problem 7: Serial correlation in a regression

Suppose the true time-series regression is

$$Y_t = \beta_0 + \beta_1 X_t + u_t, \quad u_t = \rho u_{t-1} + e_t,$$

where e_t is white noise and $|\rho| < 1$.

- (a) If $\mathbb{E}(u_t \mid X_1, \dots, X_T) = 0$, is OLS for β_1 consistent?
- (b) If $\rho \neq 0$, are the usual homoskedastic OLS standard errors generally valid?
- (c) Name one standard way to conduct inference robust to serial correlation.
- (d) Explain the difference between inconsistency of $\hat{\beta}_1$ and invalid standard errors.

Solution

Problem 8: Distributed lags and long-run multipliers

Consider the autoregressive distributed lag model

$$Y_t = \alpha + \lambda Y_{t-1} + \delta_0 X_t + \delta_1 X_{t-1} + u_t, \quad |\lambda| < 1.$$

- (a) What is the immediate effect of a one-unit increase in X_t holding Y_{t-1} fixed?
- (b) Derive the long-run multiplier of a permanent one-unit increase in X .
- (c) Compute the long-run multiplier when $\lambda = 0.5$, $\delta_0 = 0.2$, and $\delta_1 = 0.1$.
- (d) Why is the long-run multiplier different from δ_0 ?

Solution

Solution 1: Basic time-series language

Answer

- (a) A stochastic process is the probabilistic sequence $\{Y_t\}$ that could generate many possible histories. A realization is the single path we observe in the data.
- (b) The lag operator satisfies $LY_t = Y_{t-1}$. Therefore

$$\Delta Y_t = Y_t - Y_{t-1} = (1 - L)Y_t.$$

- (c) Since

$$\Delta \log P_t = \log P_t - \log P_{t-1} = \log(P_t/P_{t-1}),$$

it is approximately the proportional growth rate of the price index. For small changes, it is close to $(P_t - P_{t-1})/P_{t-1}$.

Solution 2: Weak stationarity

Moments

$$\mathbb{E}(Y_t) = \mu.$$

Because ε_t and ε_{t-1} are uncorrelated,

$$\text{Var}(Y_t) = \text{Var}(\varepsilon_t + 0.5\varepsilon_{t-1}) = \sigma^2 + 0.25\sigma^2 = 1.25\sigma^2.$$

Next,

$$\text{Cov}(Y_t, Y_{t-1}) = \text{Cov}(\varepsilon_t + 0.5\varepsilon_{t-1}, \varepsilon_{t-1} + 0.5\varepsilon_{t-2}) = 0.5\sigma^2.$$

For lag two,

$$\text{Cov}(Y_t, Y_{t-2}) = 0.$$

Conclusion

The mean is constant, the variance is constant, and autocovariances depend only on the lag. So this MA(1) process is weakly stationary.

Solution 3: Reading autocorrelations

Classification

- (a) Series A looks closest to white noise because its autocorrelations are near zero.
- (b) Series B looks persistent but plausibly stationary: the ACF is positive and decays.
- (c) Series C raises concern about a unit root or strong trend: the ACF stays very close to one and decays slowly.
- (d) The sample ACF is only a diagnostic because finite samples, structural breaks, deterministic trends, seasonality, and nonlinear dynamics can produce similar-looking patterns.

Econometric discipline

The ACF tells us what to investigate next; it is not a substitute for a maintained model, plots, transformations, and formal tests.

Solution 4: AR(1) mean, variance, and ACF

Mean

$$\mu = c + \phi\mu \quad \Rightarrow \quad \mu = \frac{c}{1 - \phi}.$$

Variance and autocorrelation

$$Y_t - \mu = \phi(Y_{t-1} - \mu) + \varepsilon_t.$$

$$\gamma_0 = \phi^2 \gamma_0 + \sigma^2 \quad \Rightarrow \quad \gamma_0 = \frac{\sigma^2}{1 - \phi^2}.$$

For $h \geq 0$,

$$\gamma_h = \phi^h \gamma_0, \quad \rho_h = \frac{\gamma_h}{\gamma_0} = \phi^h.$$

Interpretation

As $\phi \rightarrow 1^-$, shocks die out slowly and $\text{Var}(Y_t)$ becomes large.

Solution 5: AR(1) forecasting

Calculations

The unconditional mean is

$$\mu = \frac{1}{1 - 0.6} = 2.5.$$

The one-step forecast is

$$\hat{Y}_{T+1|T} = 1 + 0.6Y_T = 1 + 0.6(4) = 3.4.$$

The two-step forecast is

$$\hat{Y}_{T+2|T} = 1 + 0.6\hat{Y}_{T+1|T} = 1 + 0.6(3.4) = 3.04.$$

Mean reversion

In general,

$$\hat{Y}_{T+h|T} = \mu + 0.6^h(Y_T - \mu).$$

Since $0.6^h \rightarrow 0$, the forecast converges to $\mu = 2.5$.

Solution 6: Unit root versus trend stationarity

Classification and transformations

- (a) Model A is trend-stationary: after removing $\alpha + \delta t$, the residual u_t is stationary. Model B is difference-stationary: first differences satisfy $\Delta Y_t = g + \varepsilon_t$.
- (b) For A, detrend by regressing on a constant and time trend, then model the residual. For B, first-difference the series.
- (c) In A, a one-time shock affects u_t but dies out at rate ρ^h . In B, a one-time shock permanently shifts the level of Y_t .
- (d) If a unit-root process is incorrectly detrended, one may mistake permanent shocks for temporary cycles. If a trend-stationary process is incorrectly differenced, one may throw away useful long-run level information.

Solution 7: Serial correlation in a regression

Answer

- (a) Yes, under strict exogeneity $\mathbb{E}(u_t \mid X_1, \dots, X_T) = 0$ and standard regularity conditions, OLS is consistent even if u_t is serially correlated.
- (b) No. Serial correlation generally invalidates the usual OLS standard errors because those formulas treat regression errors as uncorrelated across time.
- (c) Use heteroskedasticity-and-autocorrelation consistent standard errors, such as Newey–West, or explicitly model the serial correlation when the maintained assumptions justify it.
- (d) Inconsistency means $\hat{\beta}_1$ does not converge to β_1 . Invalid standard errors mean the estimator may be centered correctly, but the reported uncertainty, t statistics, and confidence intervals are wrong.

Solution 8: Distributed lags and long-run multipliers

Immediate and long-run effects

- (a) Holding Y_{t-1} fixed, the immediate effect of X_t is δ_0 .
- (b) In a long-run steady state, set $Y_t = Y_{t-1} = Y$ and $X_t = X_{t-1} = X$:

$$Y = \alpha + \lambda Y + (\delta_0 + \delta_1)X.$$

Hence

$$\frac{\partial Y}{\partial X} = \frac{\delta_0 + \delta_1}{1 - \lambda}.$$

- (c) With $\lambda = 0.5$, $\delta_0 = 0.2$, and $\delta_1 = 0.1$,

$$\frac{0.2 + 0.1}{1 - 0.5} = 0.6.$$

- (d) The long-run multiplier includes the direct lag effect δ_1 and the propagation through lagged Y governed by λ .